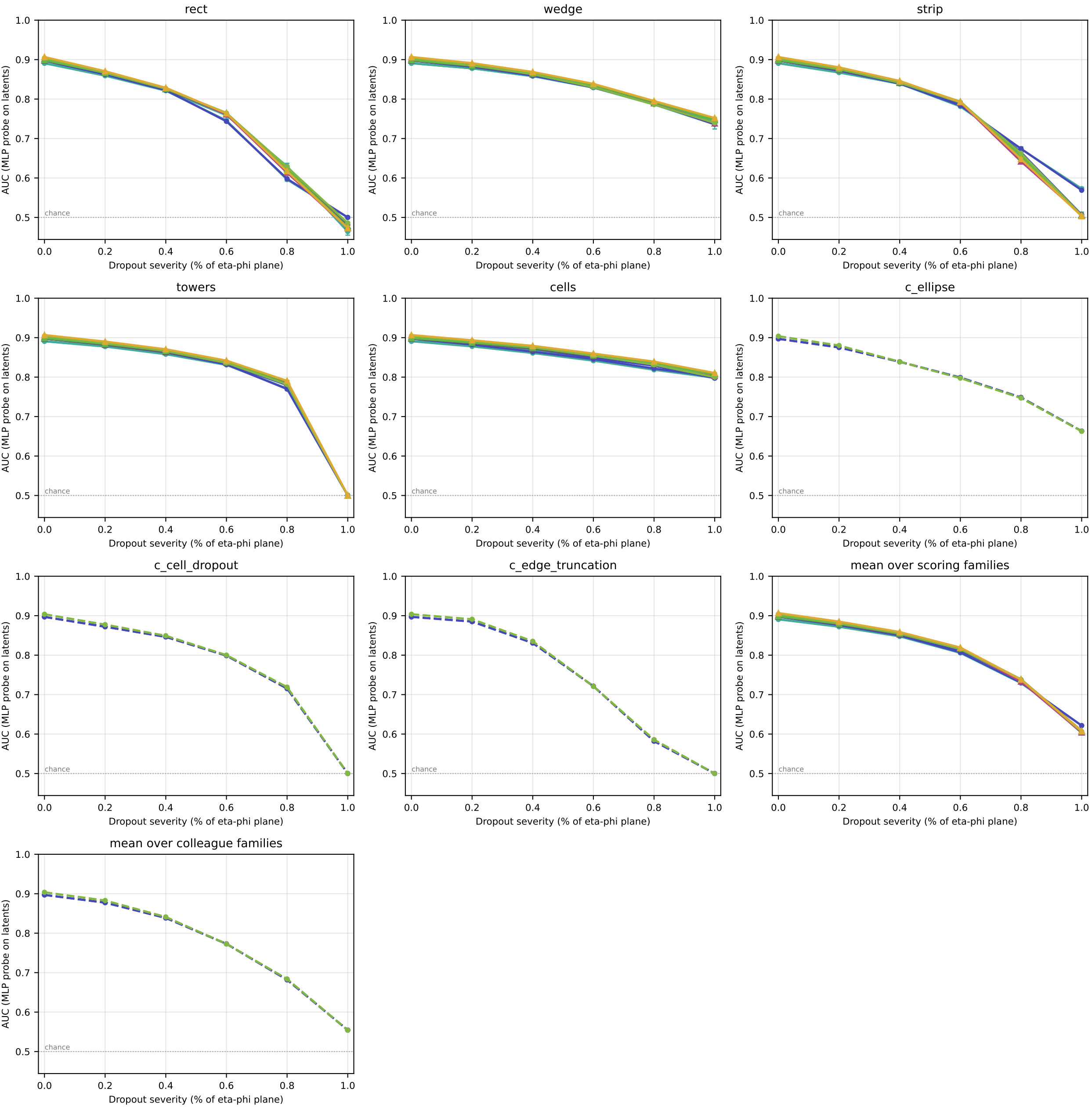


AUC vs severity on robust_tagging_eval_small_l1t.pt (L1T acceptance, eta in [-3, 3]) — not comparable to any area measured on the standard eval file



tag area over its own family set (scoring rows: mean $\pm$ std; colleague rows: per family)

- d-pma0-aug-meanpt-l1t (altdata) [trained on robust_tagging_train_data_small_l1t.pt, eta_max 3] 0.8117 $\pm$ 0.0003
- h-ref-small-s33-l1t (altdata) [trained on the standard file, eta_max 3] 0.8089 $\pm$ 0.0006
- gF-bits6-relu-l1t (altdata) [operating point: 6-bit cap + ReLU, 3.804e8 EBOPs, trained on robust_tagging_train_data_small_l1t.pt, eta_max 3] 0.8089 $\pm$ 0.0005
- g-stageA-l1t (altdata) [LayerNorm->BatchNorm, no quantization, trained on robust_tagging_train_data_small_l1t.pt, eta_max 3] 0.8078 $\pm$ 0.0007
- gD-relu-l1t (altdata) [activation ablation: ReLU at the operating cap, trained on robust_tagging_train_data_small_l1t.pt, eta_max 3] 0.8078 $\pm$ 0.0009
- winner-l1t-eta3 (altdata) [train file not recorded, eta_max 3] 0.8063 $\pm$ 0.0006
- winner-l1t (altdata) [train file not recorded, eta_max not recorded] 0.8046 $\pm$ 0.0008
- deepsets-meanpt-l1t-eta3 (altdata) [train file not recorded, eta_max 3] 0.8042 $\pm$ 0.0019
- deepsets-meanpt-l1t (altdata) [train file not recorded, eta_max not recorded] 0.8021 $\pm$ 0.0016
- h-ref-small-s33-l1t-colleague (altdata) [trained on the standard file, eta_max n/a] c_ellipse 0.8094 c_cell_dropout 0.7895 c_edge_truncation 0.7470
- c-winner-l1t (altdata) [train file not recorded, eta_max not recorded] c_ellipse 0.8084 c_cell_dropout 0.7863 c_edge_truncation 0.7436
- c-winner-l1t-eta3 (altdata) [train file not recorded, eta_max n/a] c_ellipse 0.8081 c_cell_dropout 0.7861 c_edge_truncation 0.7435

g-stageA-l1t is NOT a quantization result: it is R1+R2 (exact) plus LayerNorm->BatchNorm (not exact, forced -- HGQ2 has no quantized LayerNorm) plus 12 epochs of distillation. Its -0.0039 $\pm$ 0.0007 against d-pma0-aug-meanpt-l1t is the measured price of that swap, and it still sits above winner-l1t-eta3. Quantization on top of it costs +0.000088 (tied).